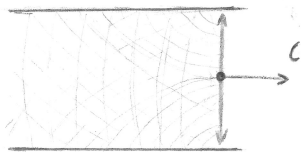


1.5 - The "Causality" Principle

Related to this discussion is the remark that, in the ultra-relativistic limit ($v \rightarrow c$), the wakes generated by a particle can only influence particles that are behind it*, since the EM fields (and thus the "wake wave-front") also propagate at speed c .

Wake analogy:



This is equivalent to saying that W must satisfy:

$$W(\vec{p}_s, \vec{p}_w, z) = 0 \quad \forall \quad z < 0 \quad !$$

Which also implies:

$$\left\{ \begin{array}{l} \bullet \vec{w}_\perp - \vec{\nabla}_{w,\perp} W = 0 \quad \forall \quad z < 0 \\ \bullet w_{||} = \frac{\partial W}{\partial z} = 0 \quad \forall \quad z < 0 \end{array} \right.$$

Even though this is a good approx. for real beams in SL storage rings, the raw effective wakes from time-domain solvers do not respect it, as they are calculated from finite-length charge distributions

The wake formalism also applies to the modelling of other sources of col. effects, such as the interaction of the beam w/ ions in the vacuum chamber, and to model the effect of CSR on the beam ~ former $\rightarrow$ causal $\checkmark$
latter $\rightarrow$ not causal !